

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/DEB_SASWI_dvw0")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3333, C2 = 6666, C3 = 1212)

text_priors <- '
Distrib (pAm,          TruncNormal_cv,          100,          0.2,          40, 2000);
Distrib (Fm,          TruncNormal,              0.1,          0.2,          0.01, 0.8);
Distrib (r_pAm_pM,    TruncNormal_cv,          0.8,          0.1,          0.1, 2);
Distrib (kap,         Uniform,                  0.1,          0.9);
Distrib (Ehp,         TruncNormal_cv,          100,          0.2,          40, 300);
Distrib (E_coc,       TruncNormal_cv,              30,          0.2, 1, 300);
Distrib (rOM_ClxFhorse, LogUniform,            0.00001,          0.3);
Distrib (f_Slope, TruncNormal_cv, 1, 1, 0, 10);
Distrib (dv, TruncNormal_cv, 2, 0.5, 0, 10);
Distrib (w0, LogUniform, 0.000000001, 0.01);
Distrib (f_EC, TruncNormal_cv, 0.5, 0.1, 0.1, 0.9);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE

# makemcsim DEBcali.model

f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcaliSWI DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	pAm.1.	Fm.1.	r_pAm_pM.1.	kap.1.	Ehp.1.	E_coc.1.
Result_mode	179.299000	0.19282900	0.37997500	0.73643600	173.33400	25.02080
2.5%	153.264000	0.16747300	0.35353500	0.68445000	147.22600	19.26160
97.5%	189.036000	0.41638300	0.51137770	0.75190400	203.21060	27.53280

Min	83.184900	0.04100250	0.30095200	0.27393000	63.11510	16.09400
Max	200.404000	0.67889400	0.81985300	0.78012800	241.75700	42.40260
Result_mean	169.566723	0.26001727	0.43013563	0.71745650	173.52660	23.13130
Result_sd	9.261653	0.06478891	0.04042193	0.01751135	14.33821	2.12655
	rOM_ClxHorse.1.	f_Slope.1.	dv.1.	w0.1.	f_EC.1.	
Result_mode	0.0019390400	0.50335200	7.2861600	1.284090e-09	0.50434700	
2.5%	0.0008198350	0.43522500	6.2977300	1.284090e-09	0.43123200	
97.5%	0.0053154900	0.66911090	9.0364400	1.307280e-04	0.61087800	
Min	0.0000212918	0.33269500	2.3433400	1.284090e-09	0.35523400	
Max	0.0671745000	2.06027000	9.9876300	1.917730e-04	0.69314100	
Result_mean	0.0027449900	0.54688399	7.6158761	5.227223e-05	0.51956541	
Result_sd	0.0012550227	0.06100233	0.7055007	4.065555e-05	0.04650688	
	Sigma_W.1.					
Result_mode	0.15039900					
2.5%	0.13411515					
97.5%	0.17235035					
Min	0.11776000					
Max	5.19852000					
Result_mean	0.15194295					
Result_sd	0.04501688					

Number of observations, n = 277

Number of parameters, k = 11

Log-likelihood, LL = -40.25135

BIC = 142.3669

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.01	1.03
E_coc.1.	1.03	1.09
Ehp.1.	1.04	1.13
f_EC.1.	1.01	1.02
f_Slope.1.	1.01	1.02
Fm.1.	1.02	1.02
kap.1.	1.02	1.04
pAm.1.	1.04	1.10
r_pAm_pM.1.	1.02	1.03
rOM_ClxHorse.1.	1.01	1.01
Sigma_W.1.	1.00	1.00
w0.1.	1.18	1.50

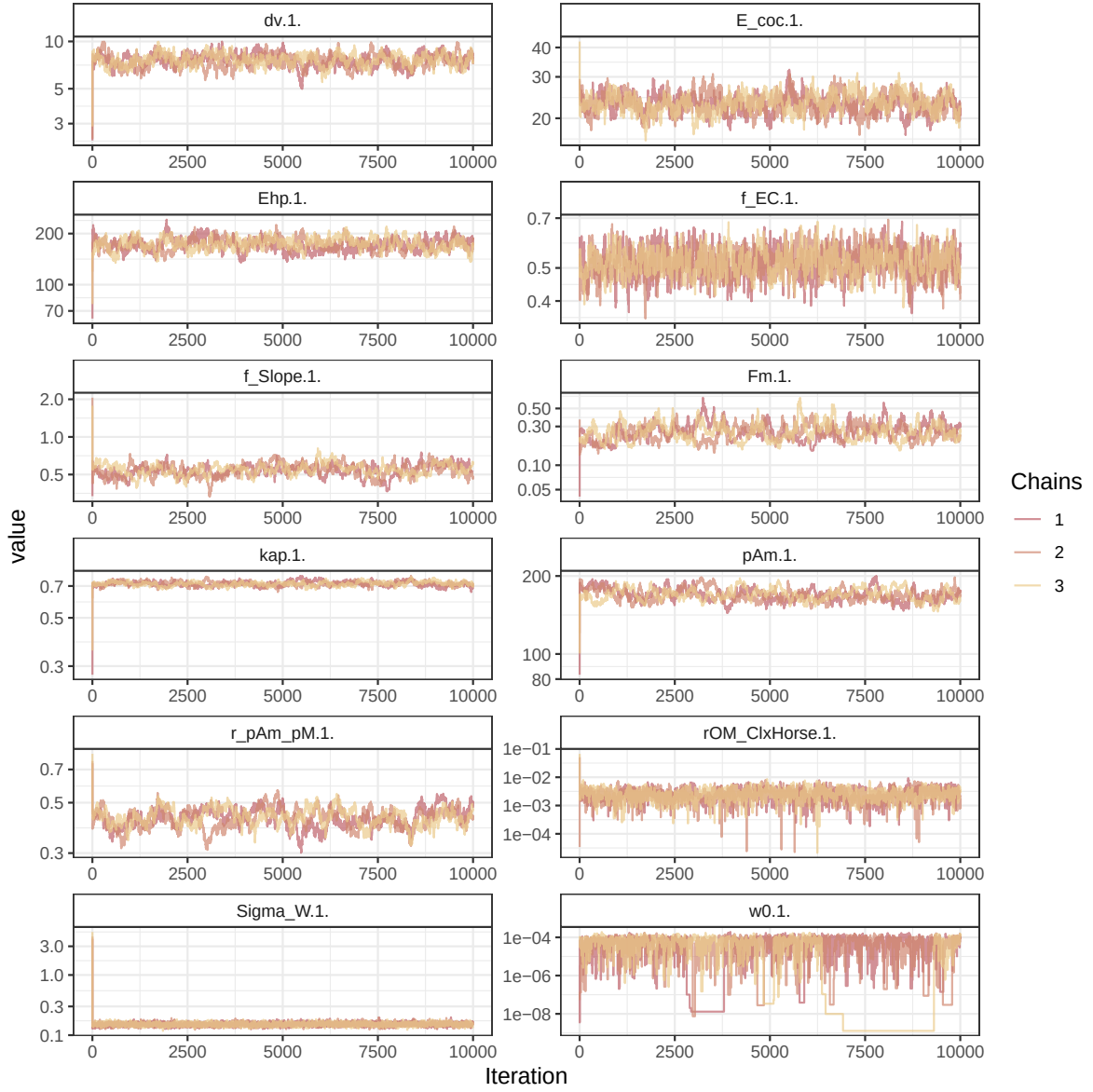


Figure 1: Parameters chains. Only the first iteration and the last $r \text{ Nb_iter_kept}$ are represented.

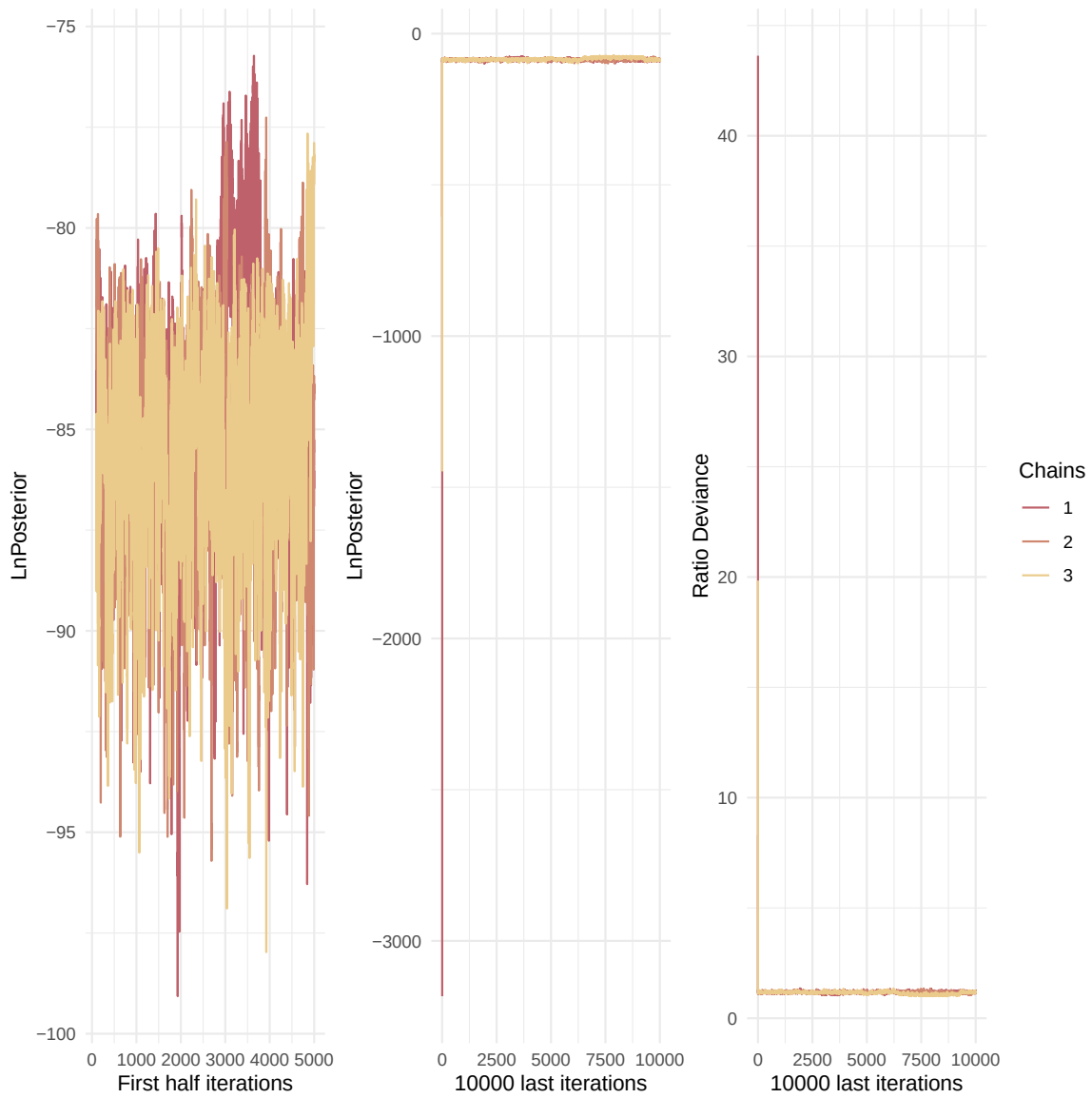


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

Multivariate psrf

1.14

```
mcmc_list_corr <- mcmc.list(
  lapply(mcmc_list, function(x) {
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc
  })
)

colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

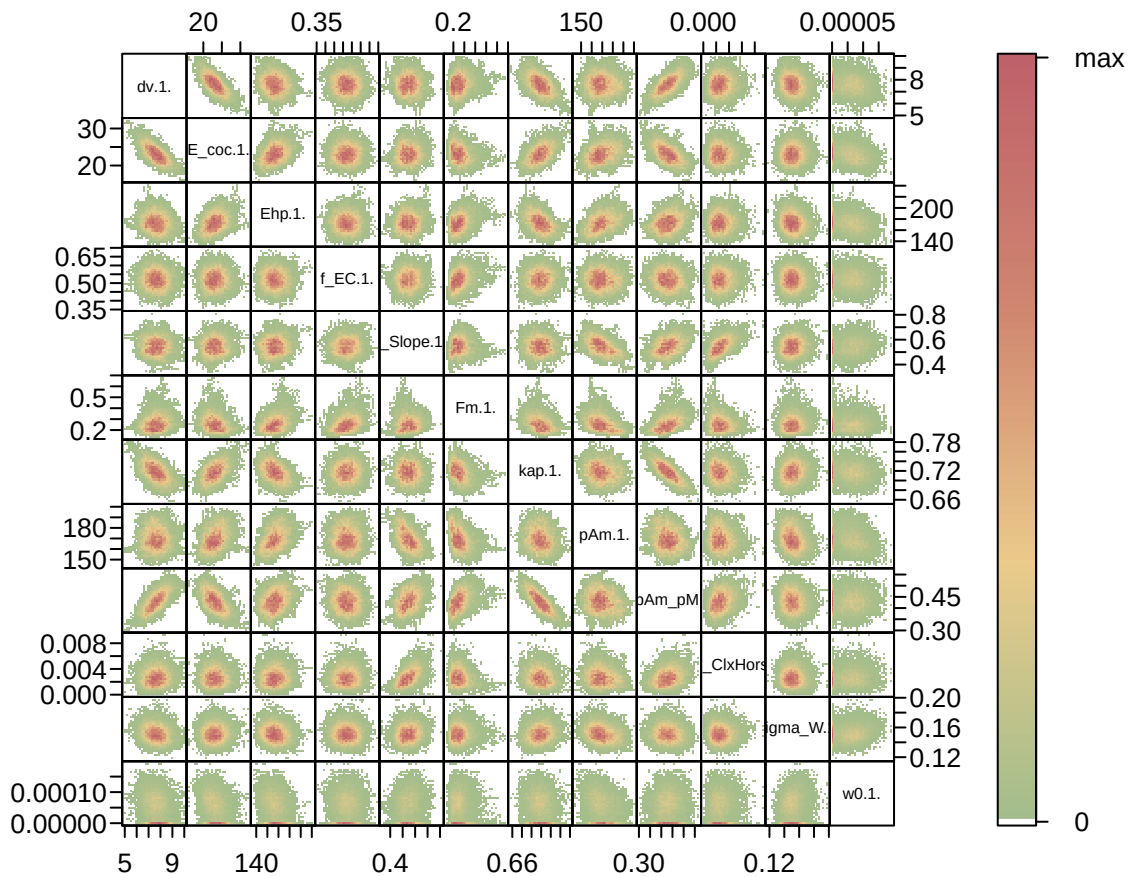


Figure 5: Parameter correlations.

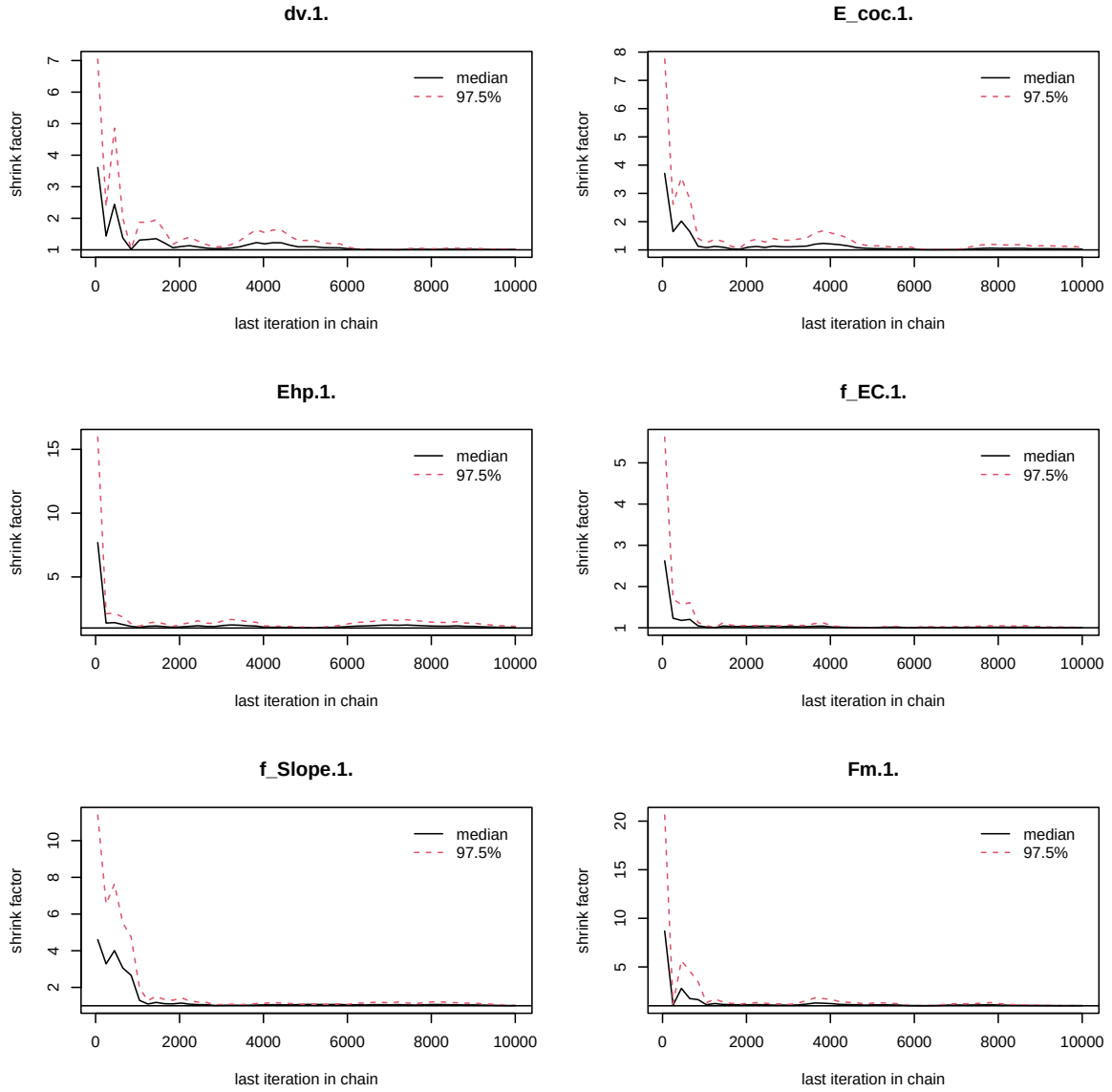


Figure 3: Chains convergence.

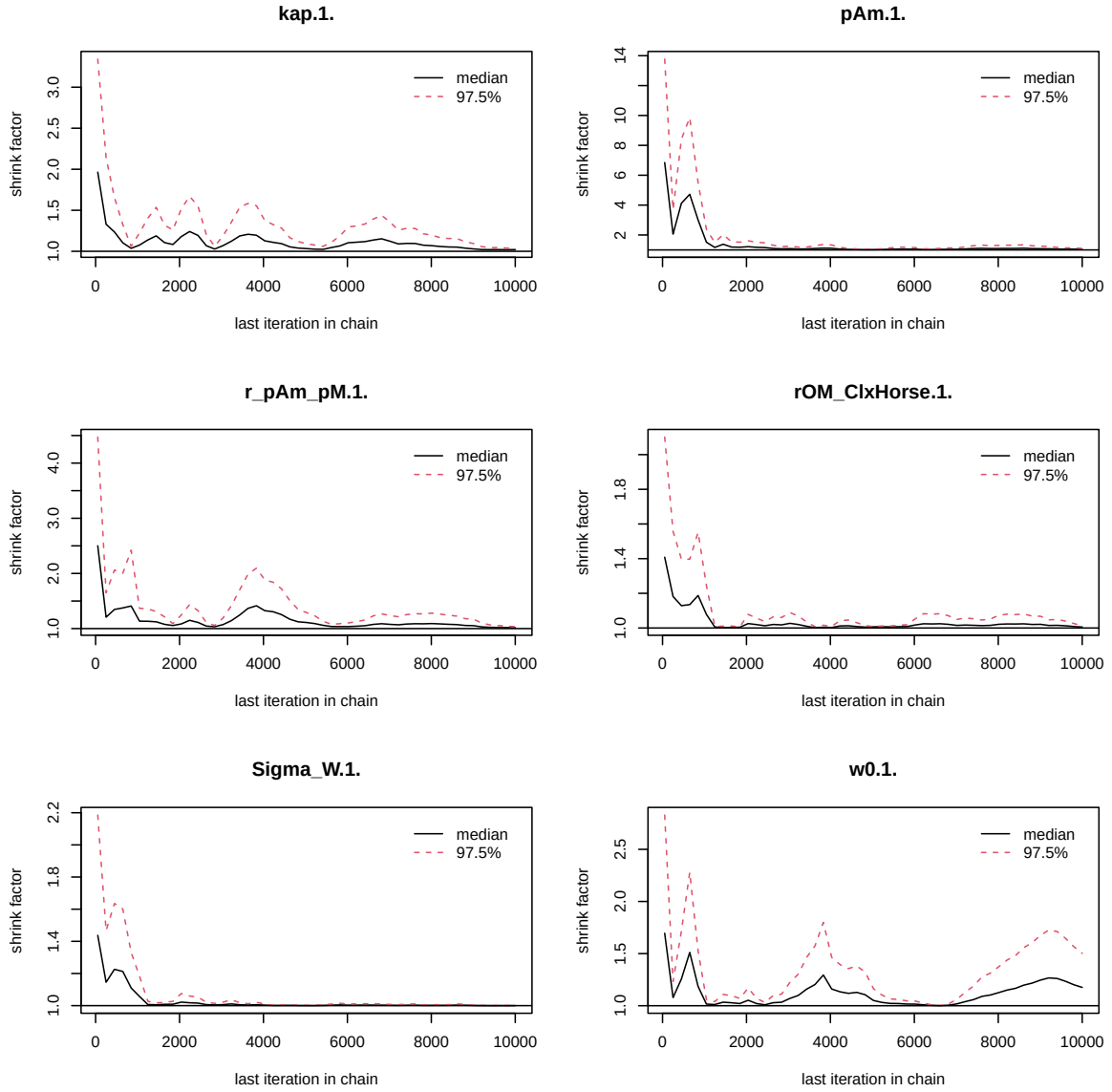


Figure 4: Chains convergence.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)

text_param <- Summary_res["Result_mode", , drop = FALSE] %>%
  t() %>%
  as.data.frame() %>%
  tibble::rownames_to_column("param") %>%
  rename(val = 2) %>%
  mutate(
    param = gsub("\\\\.1\\\. $|\\.1\\\. ", "", param),
    line = glue("{param} = {val};")
  ) %>%
  pull(line) %>%
  paste(collapse = "\n")

Endpoints_print <- "Weight,
  Length_struct,
  Reproduction,
  Energy,
  Organic_matter,"
```

Posterior
and **prior** distributions

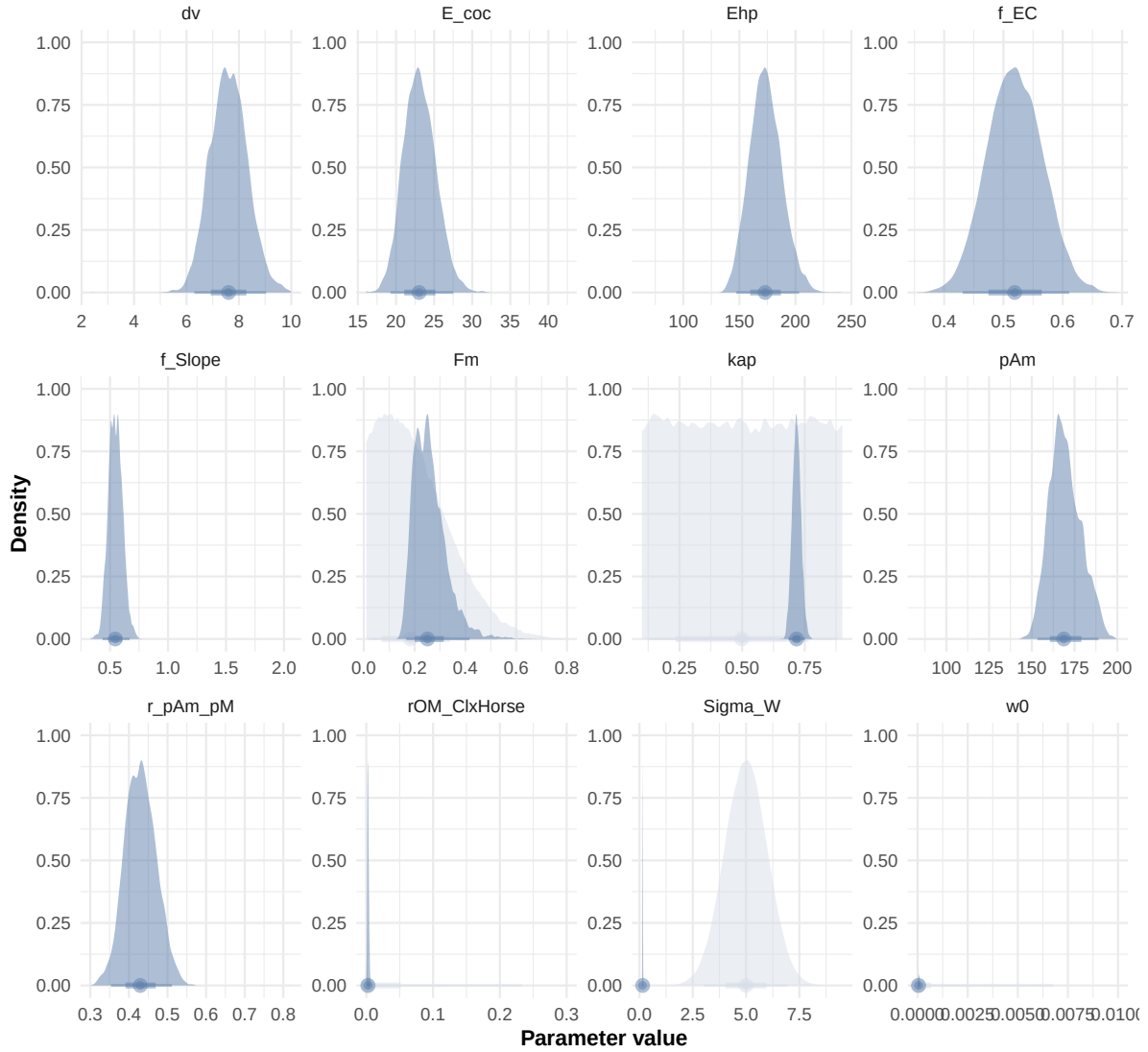


Figure 6: Distributions of the priors and the posteriors.

```
f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff, text_param, Endpoints_print)
#f_In_simulations(file_path, l_Experiments, Adults_alone, OM_diff=FALSE, text_param, text_print)

# ./mcsim.DEBcaliSWI DEB_sim_full.in
```

```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

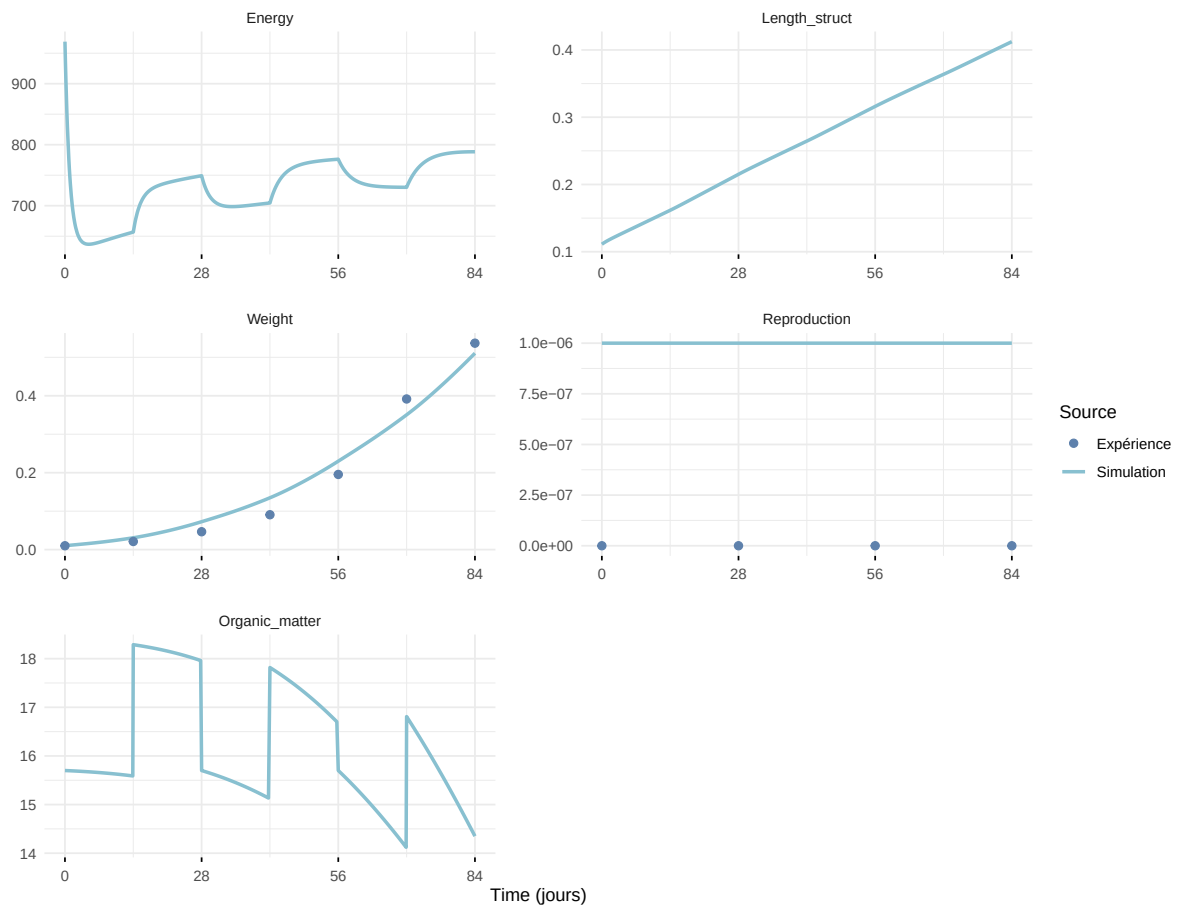
select_times <- seq(0,400, 0.1)

df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected relationship between 'x' and 'y'.
i Row 1 of 'x' matches multiple rows in 'y'.
i Row 1 of 'y' matches multiple rows in 'x'.
i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

Expérience : Bart2019_XP1_1



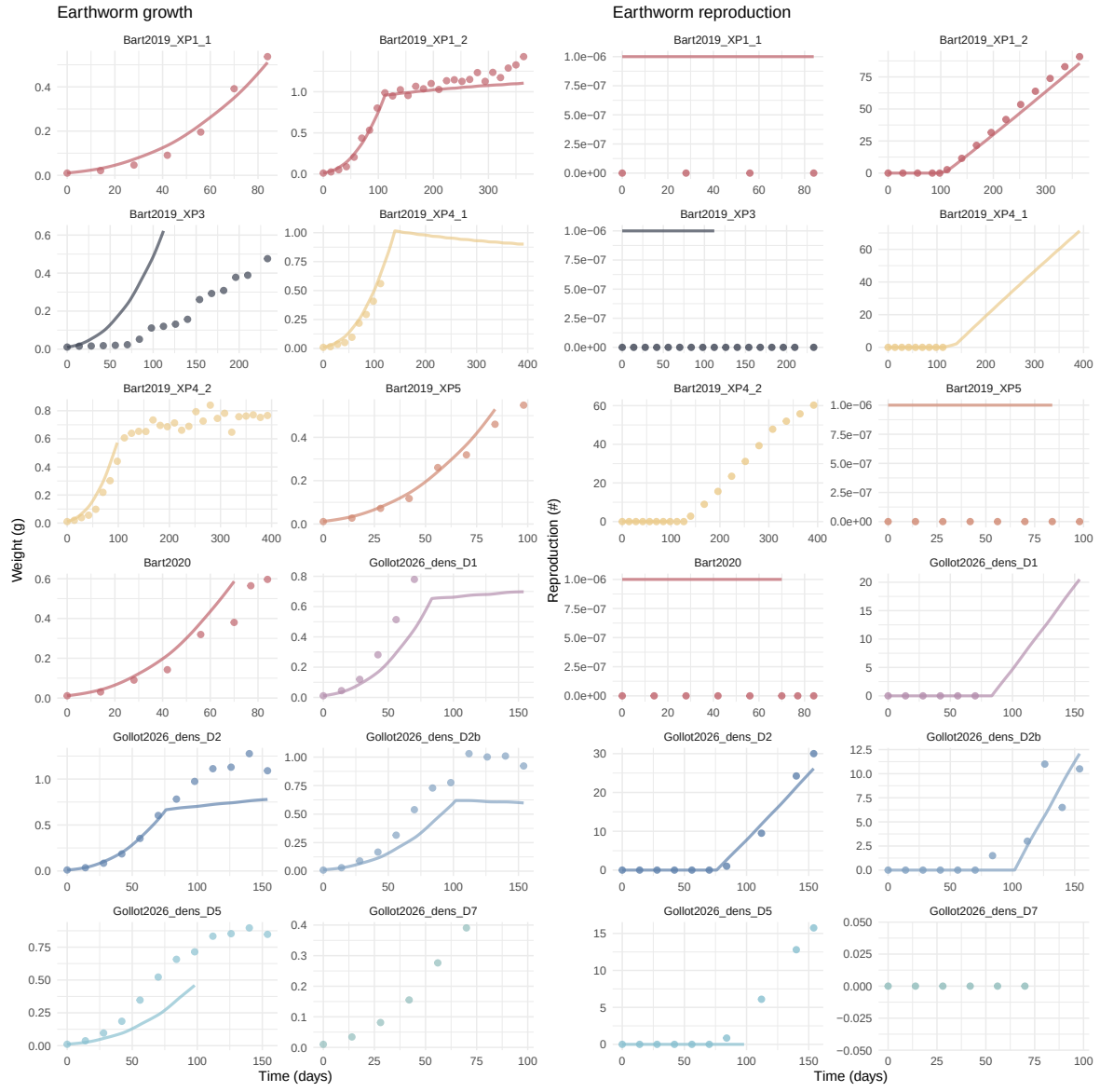
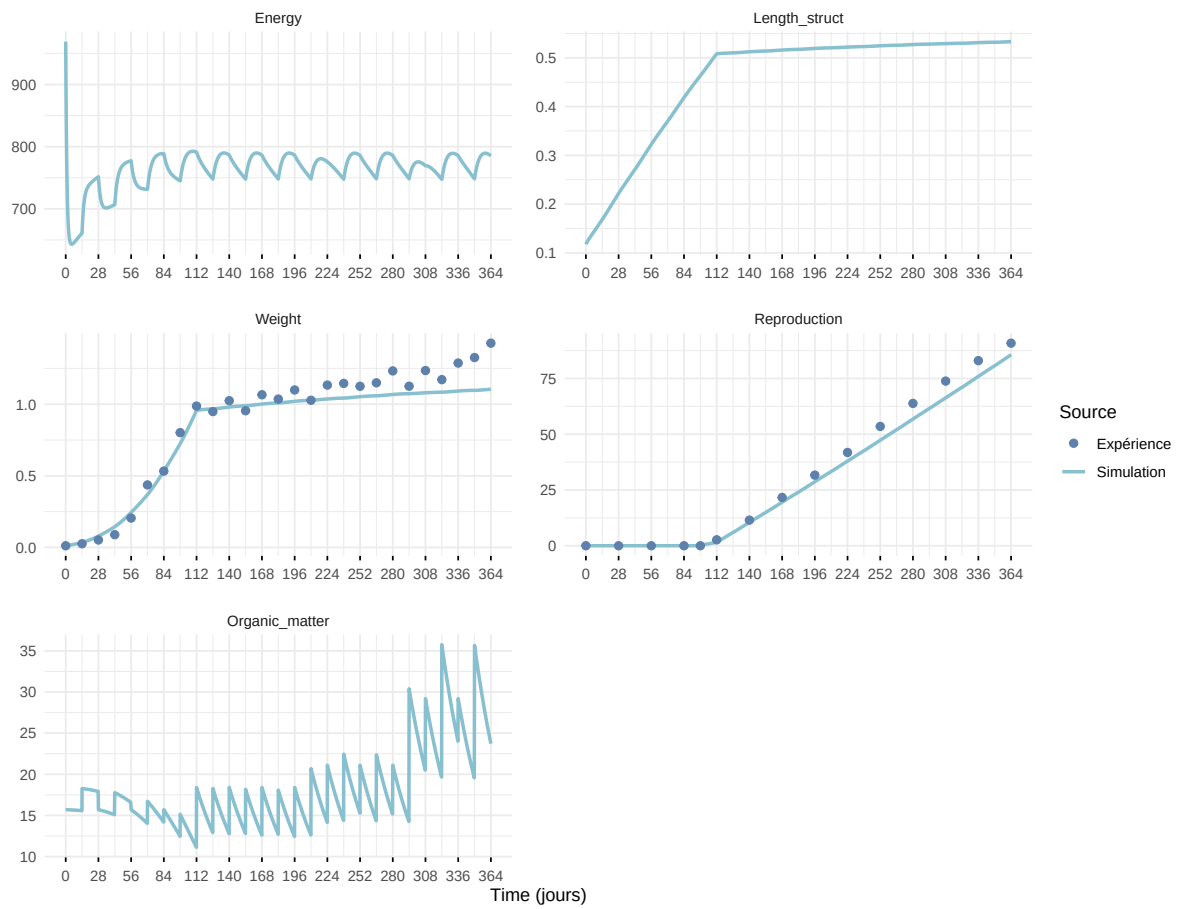
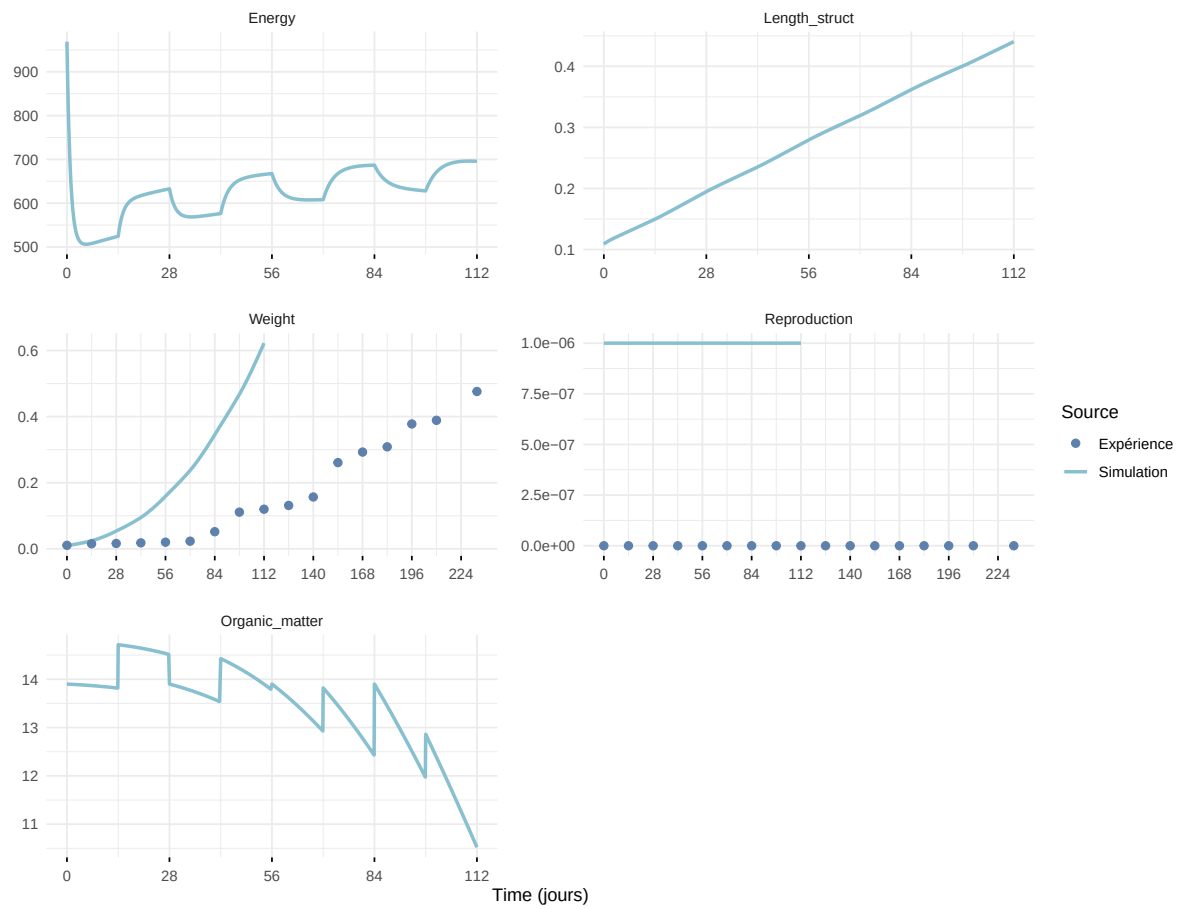


Figure 7: Simulations Weight and Reproduction.

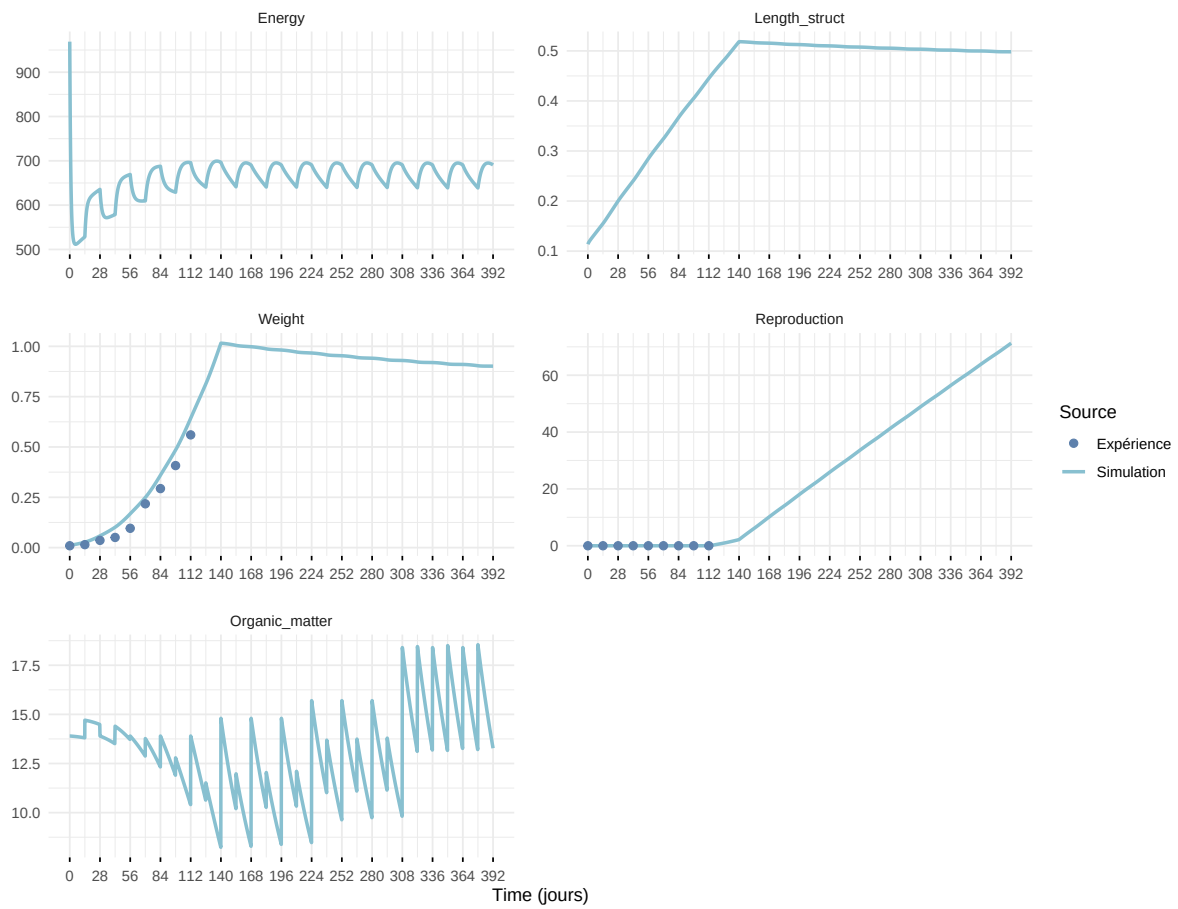
Expérience : Bart2019_XP1_2



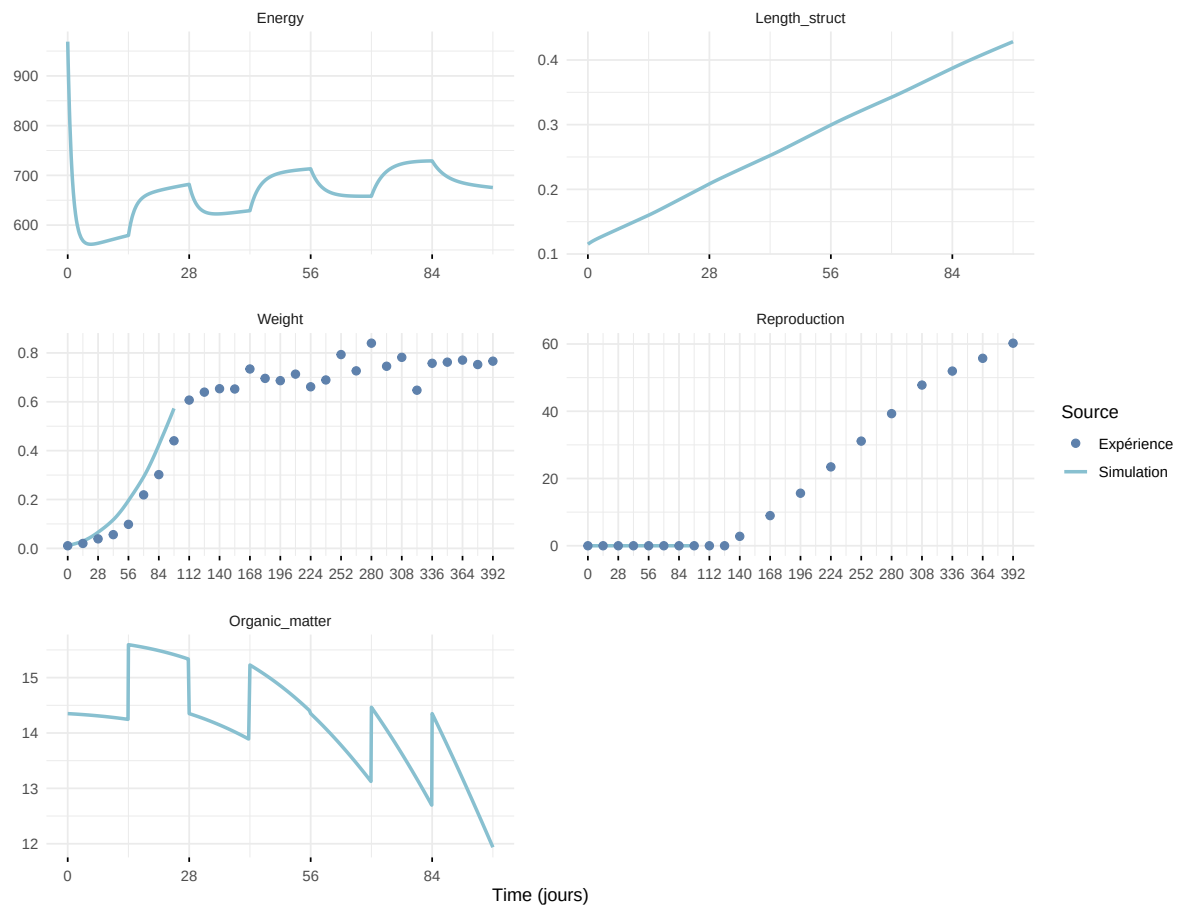
Expérience : Bart2019_XP3



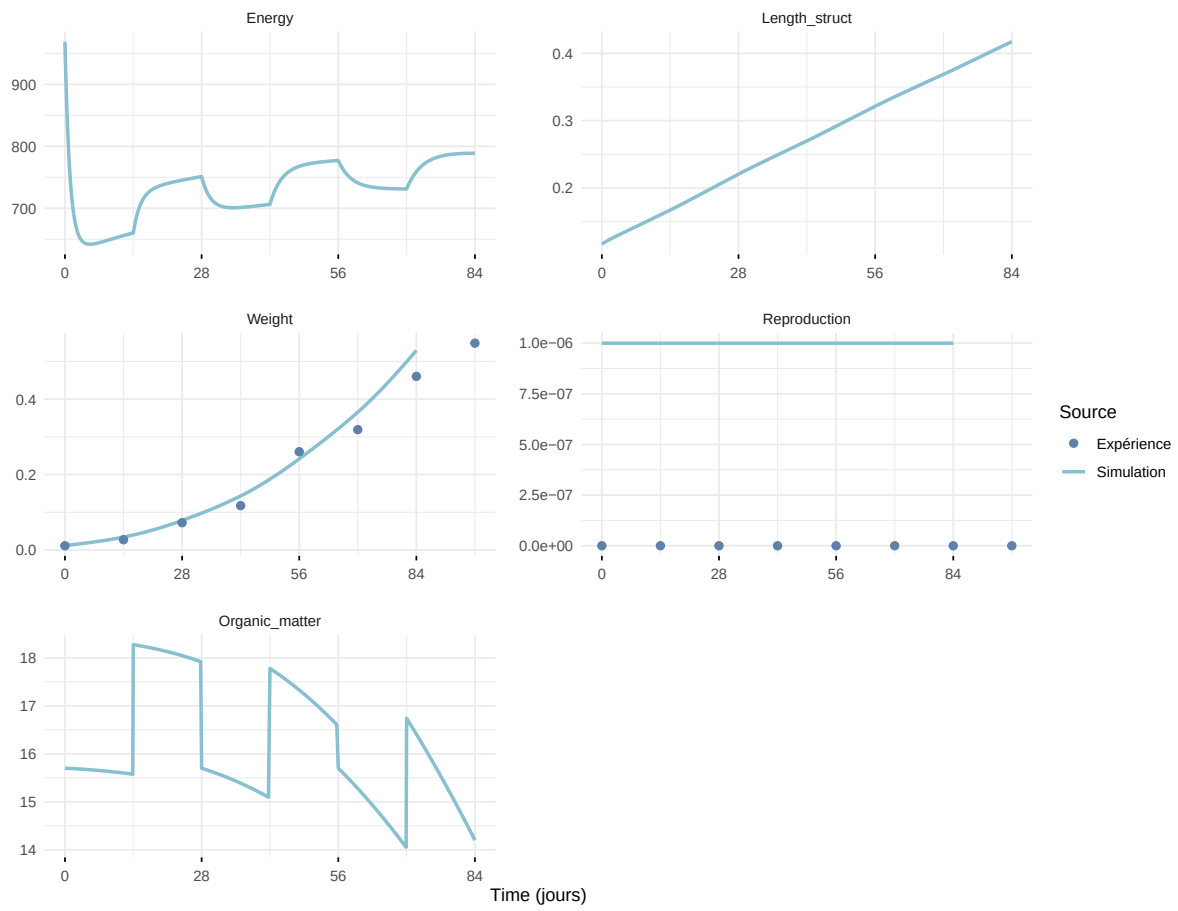
Expérience : Bart2019_XP4_1



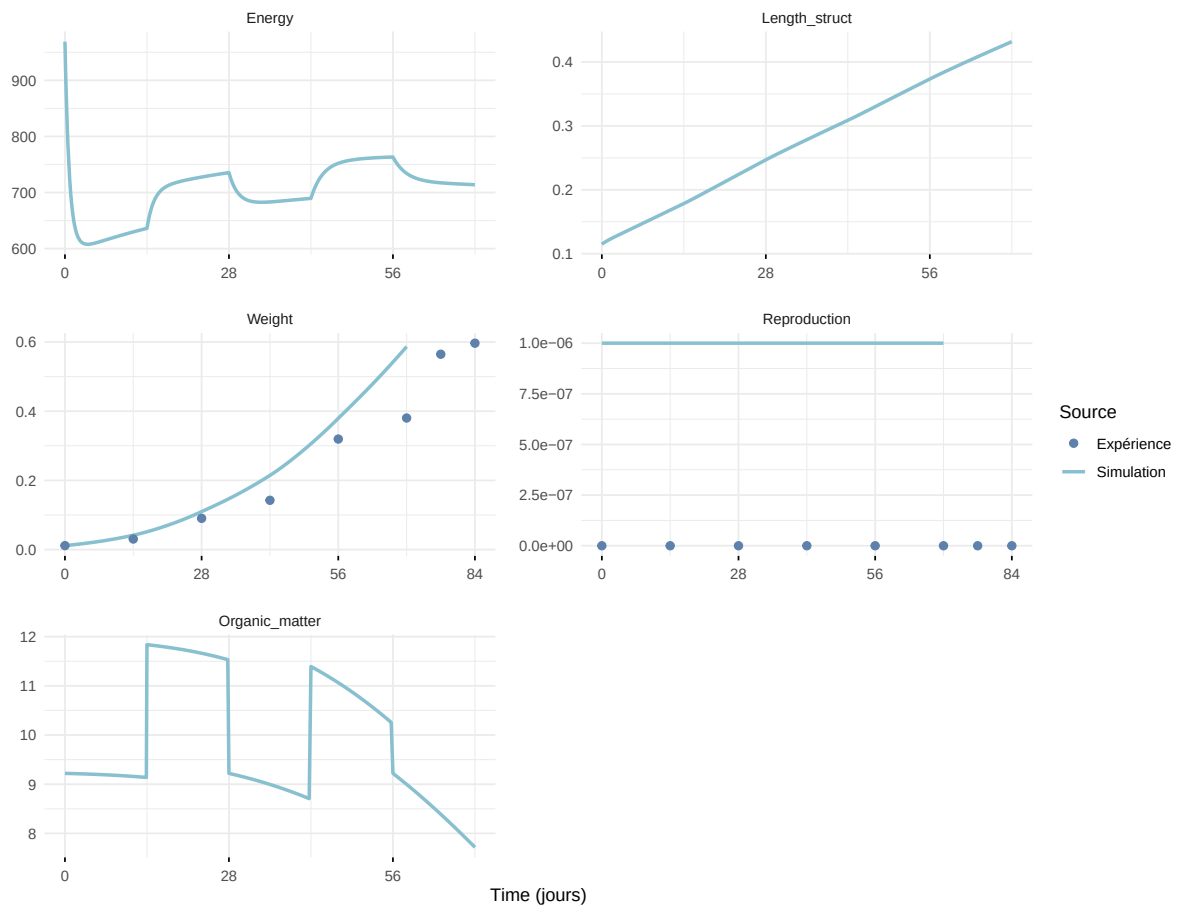
Expérience : Bart2019_XP4_2



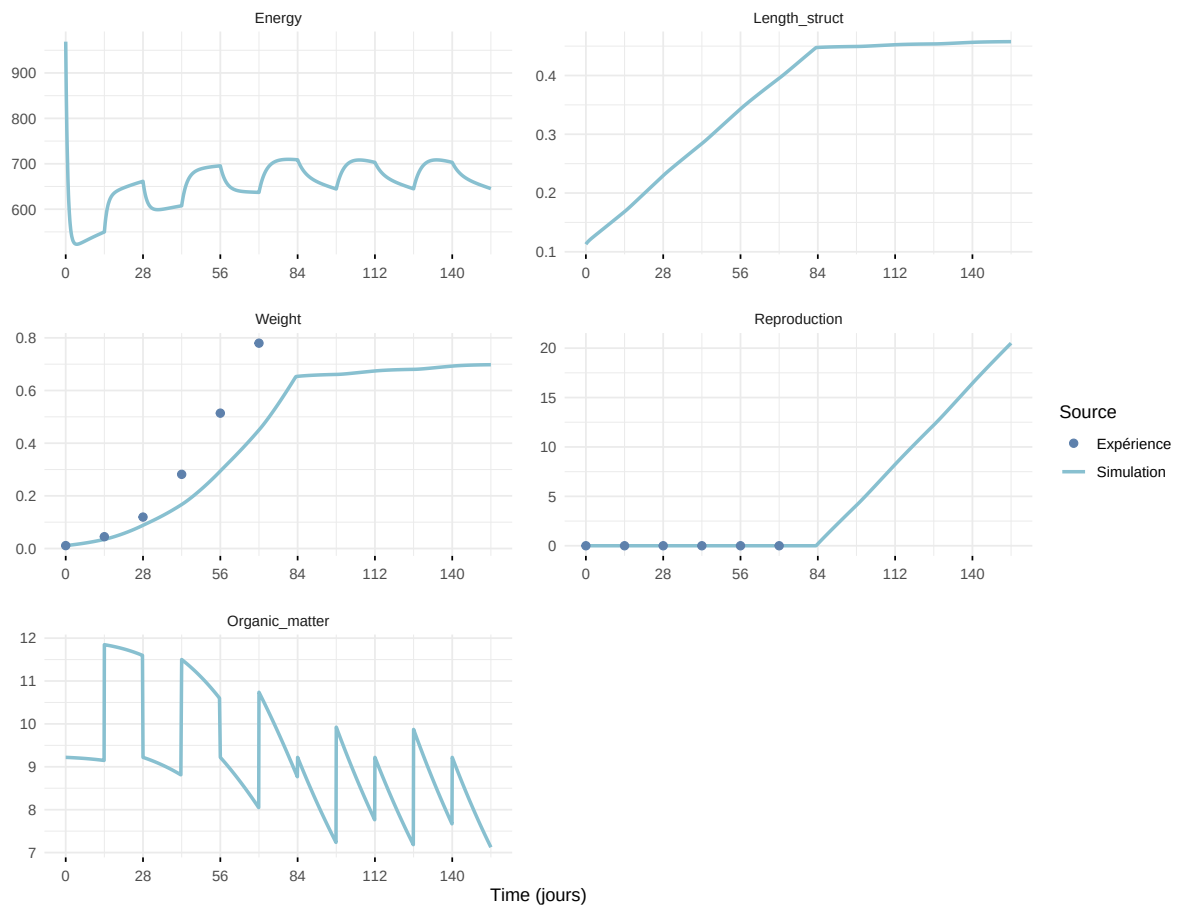
Expérience : Bart2019_XP5



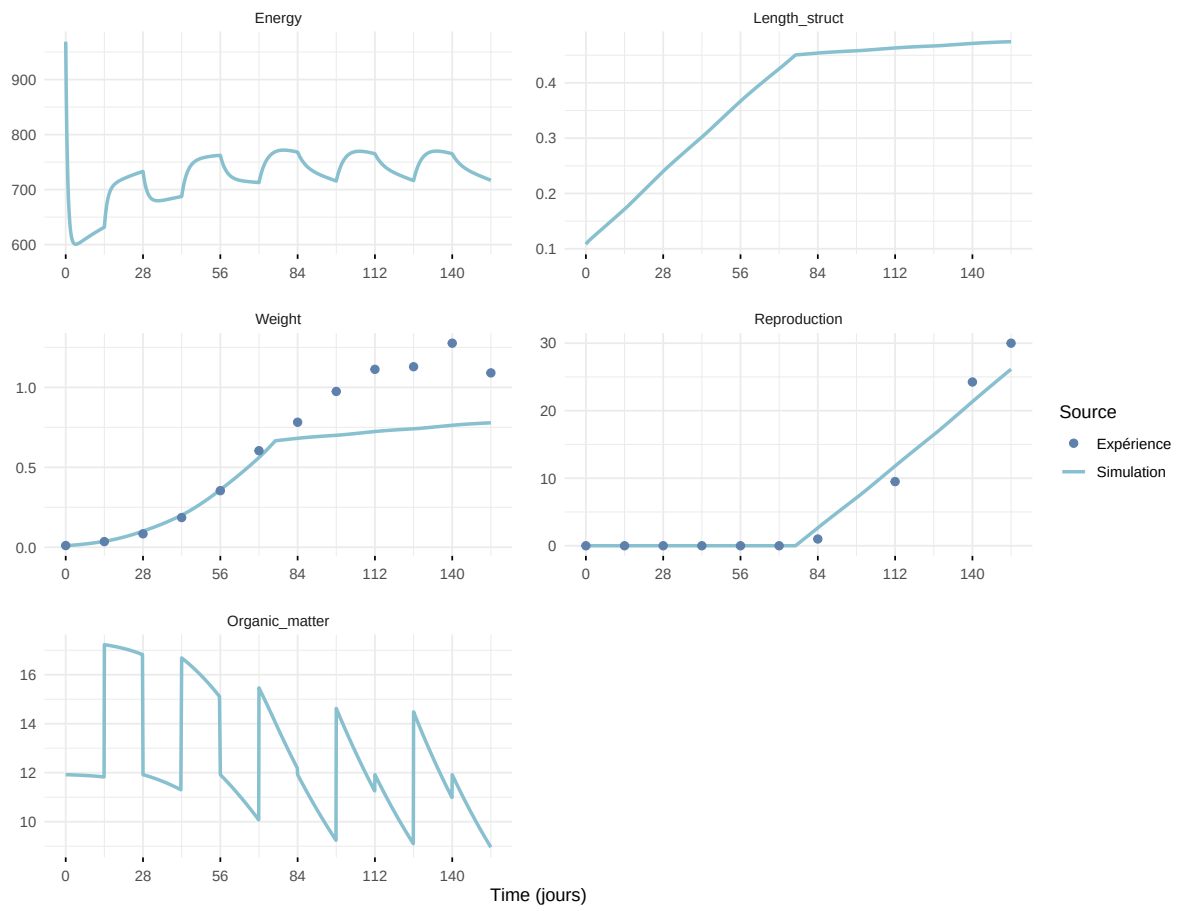
Expérience : Bart2020



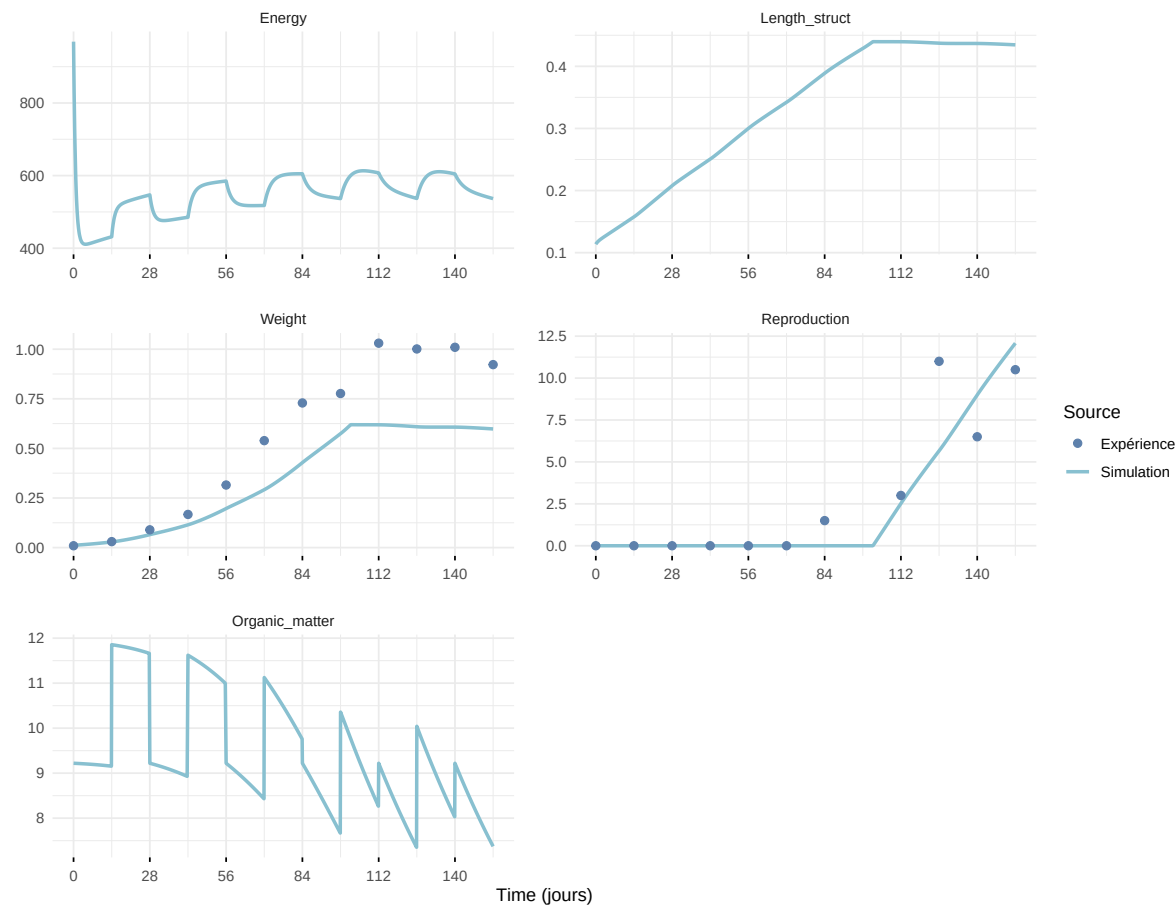
Expérience : Gollot2026_dens_D1



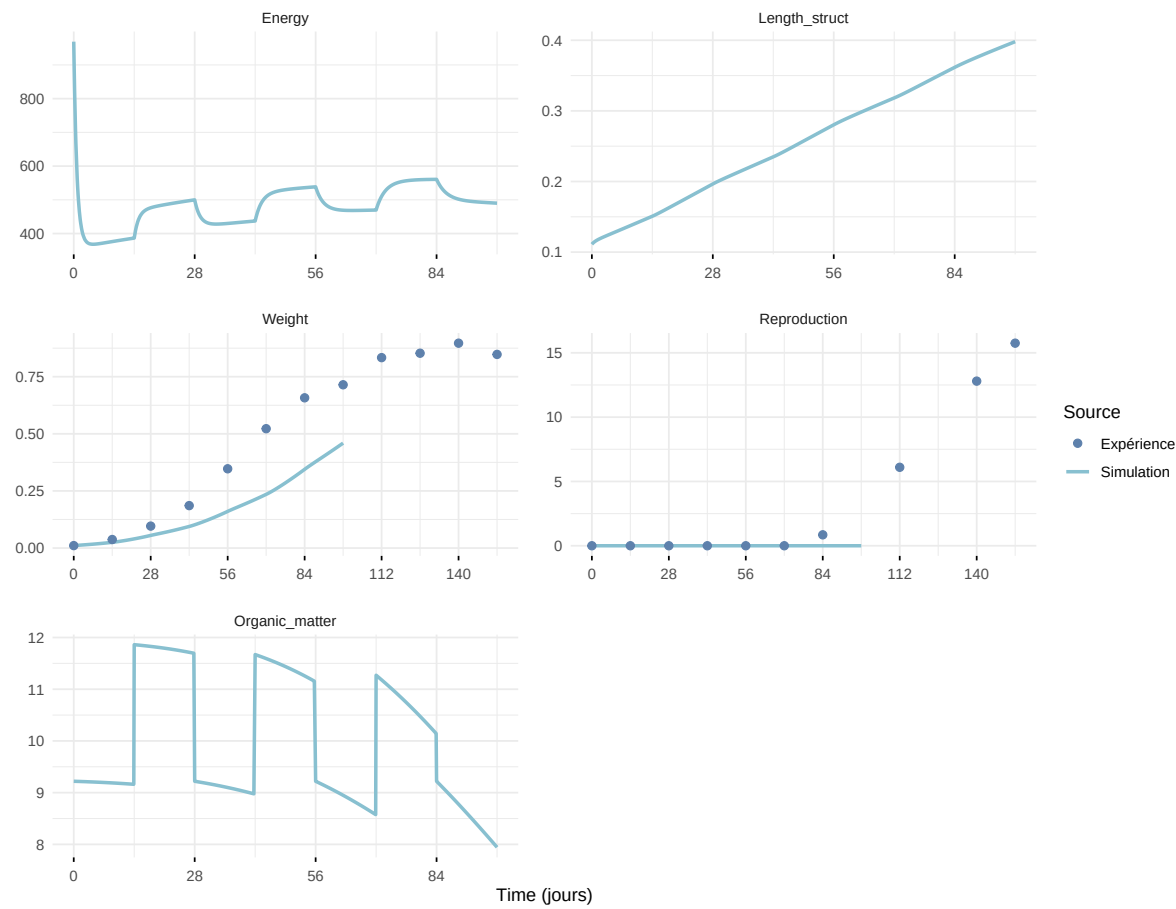
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

